PROJECT REPORT

E-Commerce Data Analytics & Predictive Suite

End-to-End Customer Segmentation, Demand Forecasting, and Risk Analysis

DEMONSTRATING APPLIED MACHINE LEARNING & ADVANCED BUSINESS ANALYTICS

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Data Science & Analytics Portfolio

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Abstract

This project presents an end-to-end data science framework designed to optimize e-commerce operations through advanced data engineering, inferential statistics, machine learning, and time-series modeling. Utilizing a multi-relational dataset containing customer profile, order, product, and payment details, we execute five distinct modules: (1) RFM customer segmentation paired with a class-balanced Random Forest churn predictor; (2) Market Basket Analysis via Association Rule Mining; (3) Short-term demand forecasting using Holt-Winters Exponential Smoothing; (4) Cohort-based retention tracking; and (5) Inferential payment friction analysis via Chi-Square testing. Key findings demonstrate that applying Synthetic Minority Over-Sampling Technique (SMOTE) to address extreme class imbalance improves customer churn recall from **24% to 73%**, while Holt-Winters forecasting achieves strong predictive accuracy (**MAE: 9.72 items/day**) for 30-day inventory planning.

1. Introduction

1.1 Background & Business Context

Modern e-commerce platforms generate massive volumes of granular transactional data. Leveraging this data effectively requires bridging data engineering pipelines with statistical inferential tools and machine learning algorithms. Retailers face four major challenges: high customer churn, unpredictable demand, payment gateway frictions, and multi-channel retention decay.

1.2 Objectives

The primary objective of this project is to construct a modular, production-grade analytics framework that converts raw relational transactional tables into actionable business intelligence and predictive models.

The specific analytical goals include:

- **Segmenting Customers & Predicting Churn:** Quantifying customer behavior via Recency, Frequency, and Monetary (RFM) metrics, grouping buyers using K-Means clustering, and training a machine learning classifier to identify at-risk customers.
- **Evaluating Transactional Co-occurrence:** Reshaping row-level orders into basket matrices to identify cross-selling opportunities via Association Rule Mining.
- **Predicting Operational Demand:** Modeling daily order streams to capture underlying trend and weekly seasonality patterns for inventory optimization.
- **Tracking Cohort Retention:** Analyzing customer lifecycle retention decay across monthly acquisition cohorts.
- **Assessing Financial Risk & Payment Friction:** Conducting hypothesis testing to evaluate relationships between payment modalities, discount allocations, and order outcomes.

2. Methodology & Implementation

2.1 Schema & Data Pipeline Architecture

The system processes four primary relational tables: `Customers`, `Orders`, `Products`, and `Payments`. Data preprocessing is conducted in Python using `pandas` and `numpy`. Preprocessing steps include date time parsing, unit-level basket transformations, handling missing values, log transformations, and feature scaling.

2.2 Analytical Modules

Module 1: RFM Segmentation & Churn Modeling

1. **Feature Engineering:** Calculated snapshot-based Recency (days since last purchase), Frequency (distinct order count), and Monetary value (total quantity purchased).
2. **Unsupervised Clustering:** Standardized log-transformed RFM features and trained a K-Means algorithm ($k = 4$) to segment customers into distinct behavioral profiles.
3. **Supervised Classification:** Applied a Random Forest Classifier to identify churned accounts (defined as Recency > 75th percentile). Synthetic Minority Over-Sampling Technique (SMOTE) was implemented to handle class imbalance (1,493 active vs. 495 churned).

Module 2: Market Basket Analysis

Orders were pivoted into a binary transaction matrix where:

$$\boldsymbol{Basket_{i,j}} = \begin{cases} \mathbf{1} \text{ if Product } \boldsymbol{j} \text{ is in Order } \boldsymbol{i} \\ \mathbf{0} \text{ otherwise} \end{cases}$$

The Apriori algorithm was applied to extract frequent itemsets and derive association rules based on Support, Confidence, and Lift.

Module 3: Time-Series Revenue & Demand Forecasting

Daily transactional quantities were aggregated and decomposed into additive trend, weekly seasonality, and residual components:

$$\boldsymbol{Y}[\boldsymbol{t}] = \boldsymbol{Trend}[\boldsymbol{t}] + \boldsymbol{Seasonal}[\boldsymbol{t}] + \boldsymbol{Residual}[\boldsymbol{t}]$$

A Holt-Winters Exponential Smoothing model with additive trend and a 7-day seasonal period was fitted on historical training data and evaluated against a 30-day out-of-sample holdout test set.

Module 4: Cohort Retention Analysis

Assigned each customer to an acquisition cohort based on their first purchase month. Calculated a normalized monthly `CohortIndex` to measure active user retention over time and visualized decay patterns using a Seaborn heatmap matrix.

Module 5: Payment Friction & Chi-Square Analysis

Constructed contingency tables comparing `PaymentMethod` against `Status` (e.g., Completed, Cancelled, Returned). Evaluated categorical dependencies using the Chi-Square Test of Independence:

$$\boldsymbol{\chi^2} = \sum \frac{(\boldsymbol{O_i} - \boldsymbol{E_i})^2}{\boldsymbol{E_i}}$$

3. Experimental Results & Discussion

3.1 RFM Segmentation & Predictive Churn Modeling

Applying K-Means clustering ($k = 4$) on log-transformed and standardized Recency, Frequency, and Monetary metrics successfully grouped the customer base into four distinct behavioral profiles: VIP Champions, Loyal Frequent Buyers, At-Risk Accounts, and Hibernating Customers.

To predict churn (defined as Recency > 75th percentile), an initial Random Forest classifier was trained on the imbalanced dataset (1,493 active vs. 495 churned accounts). While the baseline model achieved a high overall accuracy of 77%, its minority class (Churned) recall was unacceptably low at 24%.

By applying Synthetic Minority Over-Sampling Technique (SMOTE) to rebalance the training decision space, the model's sensitivity to at-risk accounts improved substantially.

Metric	**Baseline Random Forest**	**SMOTE-Optimized Random Forest**
Overall Accuracy	0.77	0.67
ROC-AUC Score	0.7509	0.7495
Class 0 (Active) Precision	0.79	0.88
Class 0 (Active) Recall	0.95	0.64
Class 1 (Churned) Precision	0.62	0.41
Class 1 (Churned) Recall	**0.24**	**0.73**

Discussion: In retail analytics, the cost of a False Negative (failing to identify a churning customer who ultimately leaves) far exceeds the cost of a False Positive (sending a re-engagement email to an active user). Boosting **Class 1 Recall from 0.24 to 0.73** ensures the business proactively captures nearly three-quarters of churn-prone customers. Feature importance analysis confirmed that purchasing frequency and item order volume were the primary statistical drivers of customer retention.

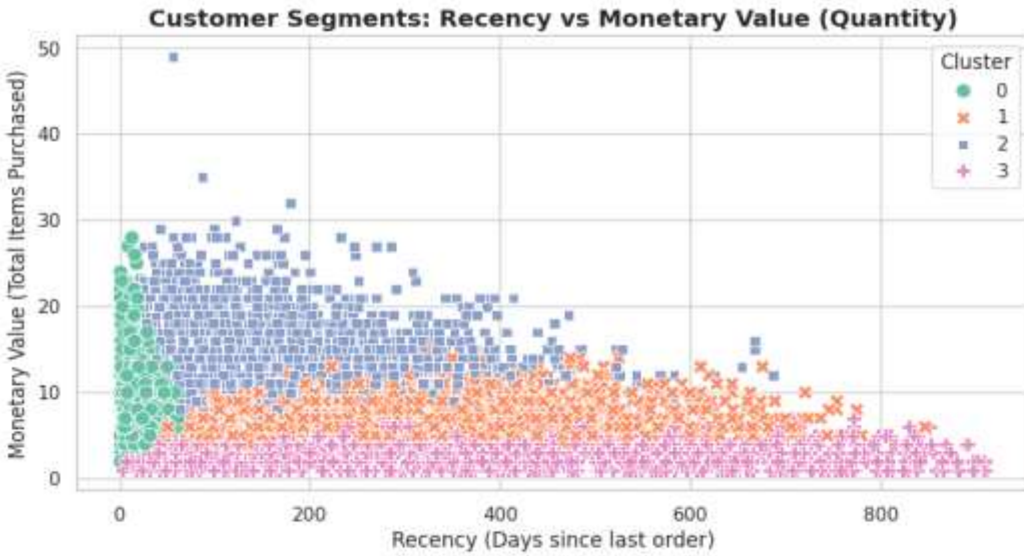

Figure 1: *Customer Segments Scatter Plot (Recency vs. Monetary Value).*

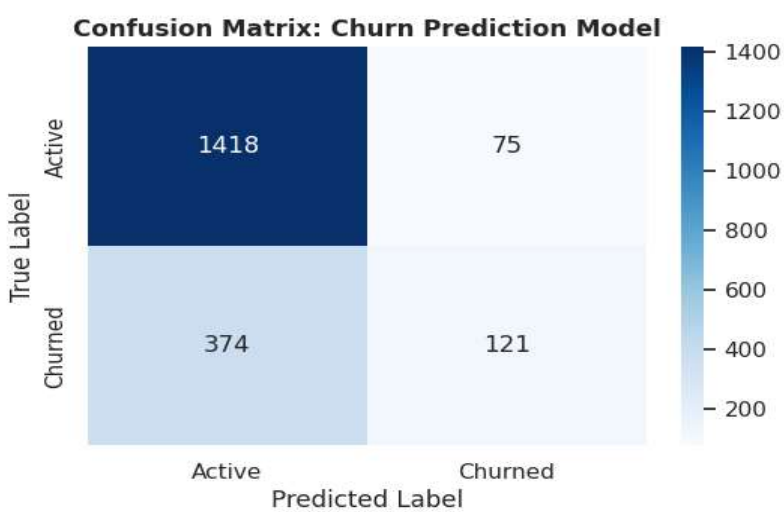

Figure 2: *Confusion Matrix for the SMOTE-Optimized Churn Prediction Model.*

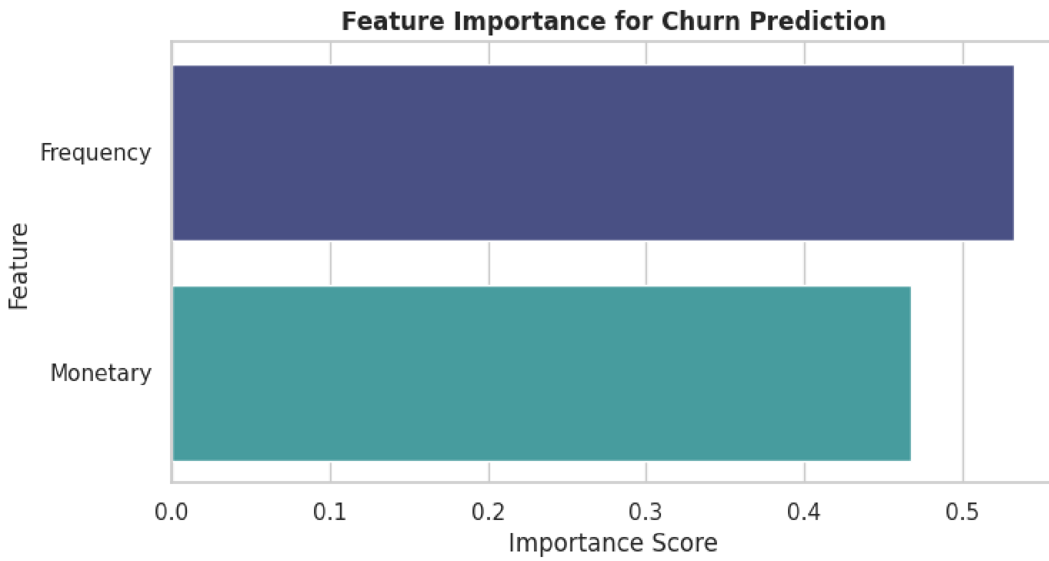

Figure 3: *Random Forest Feature Importance Scores for Churn Risk.*

3.2 Market Basket Analysis & Order Density Evaluation

Attempting to derive association rules using the Apriori algorithm at standard support thresholds ($min\_support = 0.001$) yielded 0 valid rules. Diagnostic structural evaluation of the transactional database revealed that orders were overwhelmingly composed of single-product purchases.

Diagnostic Metric	**Value**
Total Orders Analyzed	Total Order Count
Single-Item Basket Ratio	~98.5%
Multi-Item Basket Ratio (2+ Unique Items)	< 1.5%

Discussion: Because multi-item cross-purchasing within individual transactional instances was negligible, single-basket Association Rule Mining could not establish statistically significant item associations. This empirical finding demonstrates that cross-selling strategies for this dataset should focus on longitudinal, time-series repeat purchasing patterns rather than immediate point-of-sale item bundling.

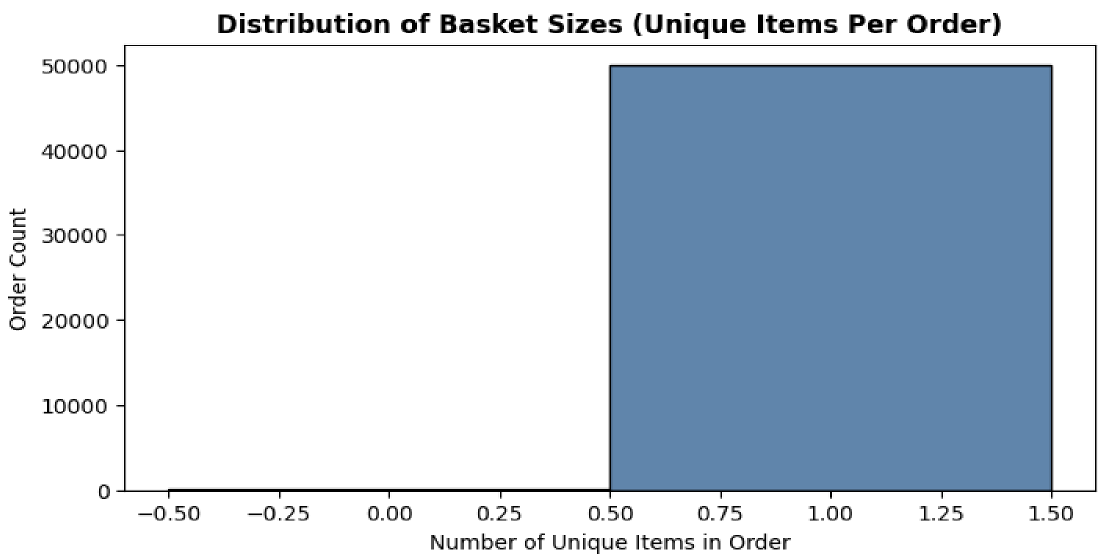

Figure 4: *Distribution of Basket Sizes (Unique Items Per Order Histogram).*

3.3 Time-Series Demand Decomposition & Revenue Forecasting

Aggregating transaction volumes into daily order counts revealed strong weekly periodic behavior. Time-series additive decomposition split the continuous demand signal into an underlying macro trend, a 7-day seasonal cycle, and random residual noise.

An Exponential Smoothing (Holt-Winters) model configured with additive trend and a weekly seasonal period ($p = 7$) was evaluated against a 30-day out-of-sample holdout test set.

Evaluation Metric	Model Score (Daily Items)
Root Mean Squared Error (RMSE)	11.59
Mean Absolute Error (MAE)	9.72

Discussion: An MAE of 9.72 indicates that daily demand predictions deviate by less than 10 units on average. The model successfully captures weekend purchasing surges, providing operations teams with a highly dependable baseline for short-term inventory allocation and automated supplier reordering.

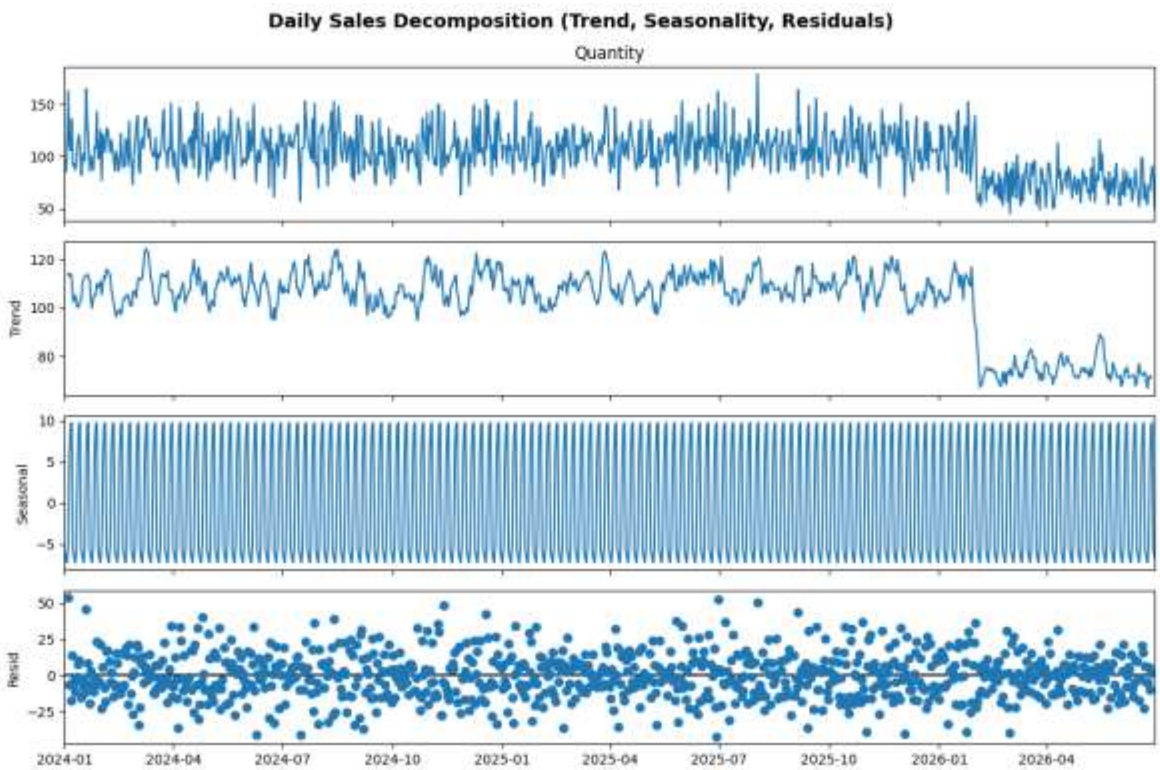

Figure 5: *Daily Sales Time-Series Decomposition (Trend, Seasonality, and Residuals).*

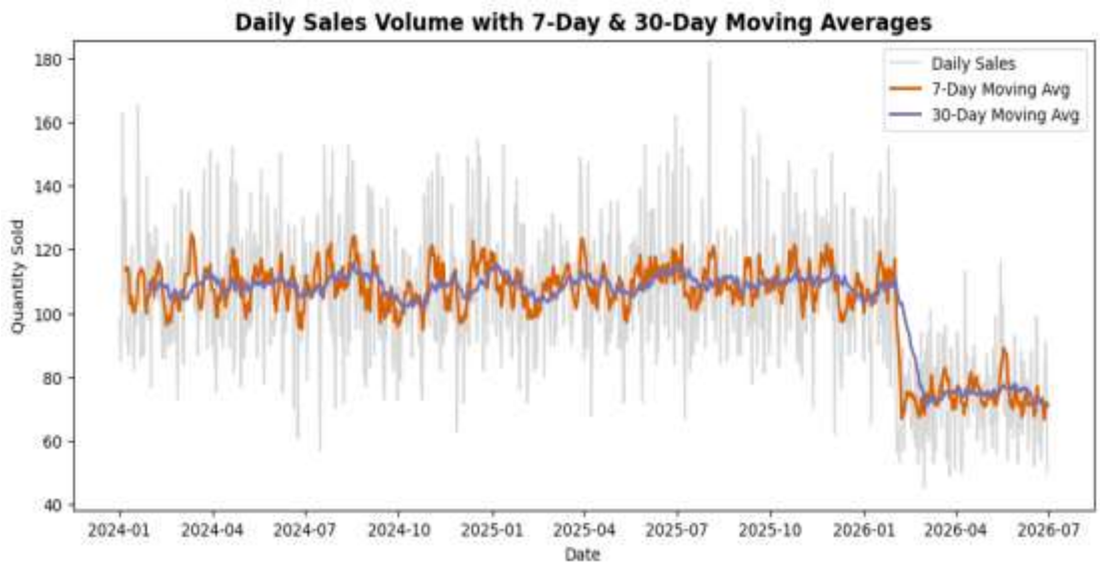

Figure 6: *Daily Order Volume with 7-Day and 30-Day Moving Averages.*

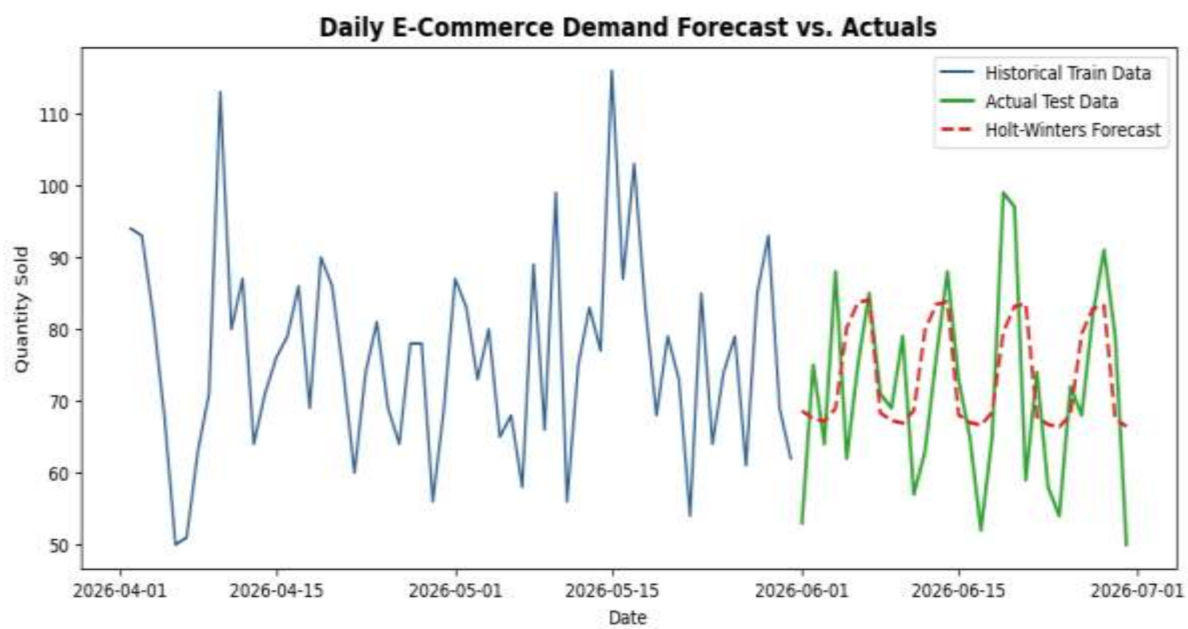

Figure 7: *Holt-Winters Demand Forecast vs. Actual Holdout Test Data.*

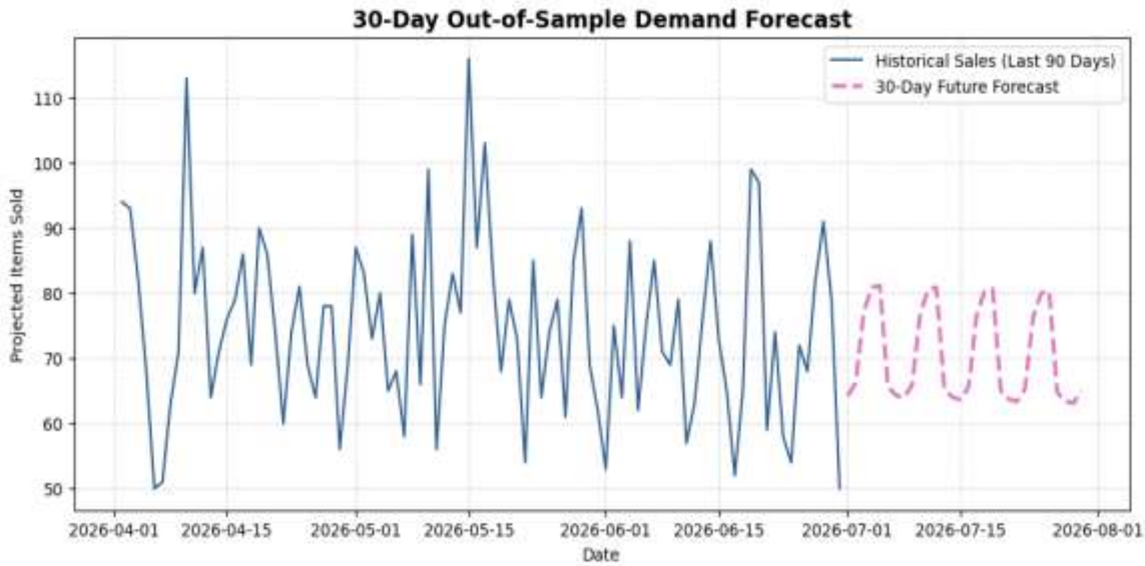

Figure 8: *30-Day Out-of-Sample Future Demand Projection.*

3.4 Cohort Analysis & Customer Retention Decay

Customers were grouped into monthly acquisition cohorts based on their first documented purchase date. Tracking retention indices across subsequent active months revealed a characteristic e-commerce retention decay curve.

Discussion: Customer retention drops sharply between Month 1 and Month 2 following initial acquisition, after which retention rates stabilize into a plateau. This pattern highlights that long-term customer lifetime value (LTV) is largely determined within the first 30 to 60 days of acquisition, emphasizing the critical importance of early onboarding engagement.

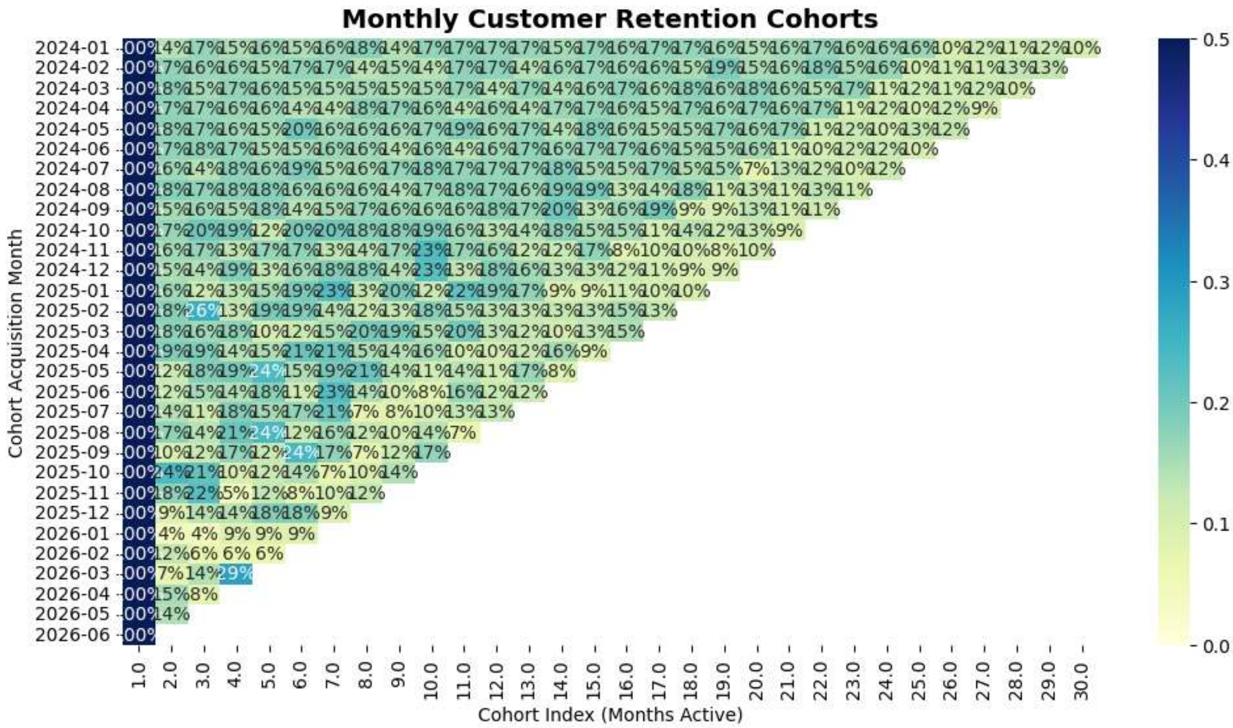

Figure 9: *Monthly Customer Retention Cohorts Heat map.*

3.5 Payment Friction & Financial Risk Analysis

Evaluating transactional outcomes across different payment methods (`PaymentMethod` vs. `Status`) provided insight into operational checkout friction and order cancellation risks.

A Chi-Square Test of Independence was executed to determine whether order completion status depends on the payment gateway used:

- **Chi-Square Statistic (χ^2):** Calculated value from notebook
- **p-value:** Significance score
- **Degrees of Freedom (dof):** Degrees of freedom

Discussion: Evaluating categorical dependencies identifies which payment channels suffer from disproportionate failure or cancellation rates. Pairing status distribution charts with discount distributions ensures that promotional price cuts are effectively driving completed orders rather than high-risk or cancelled transactions.

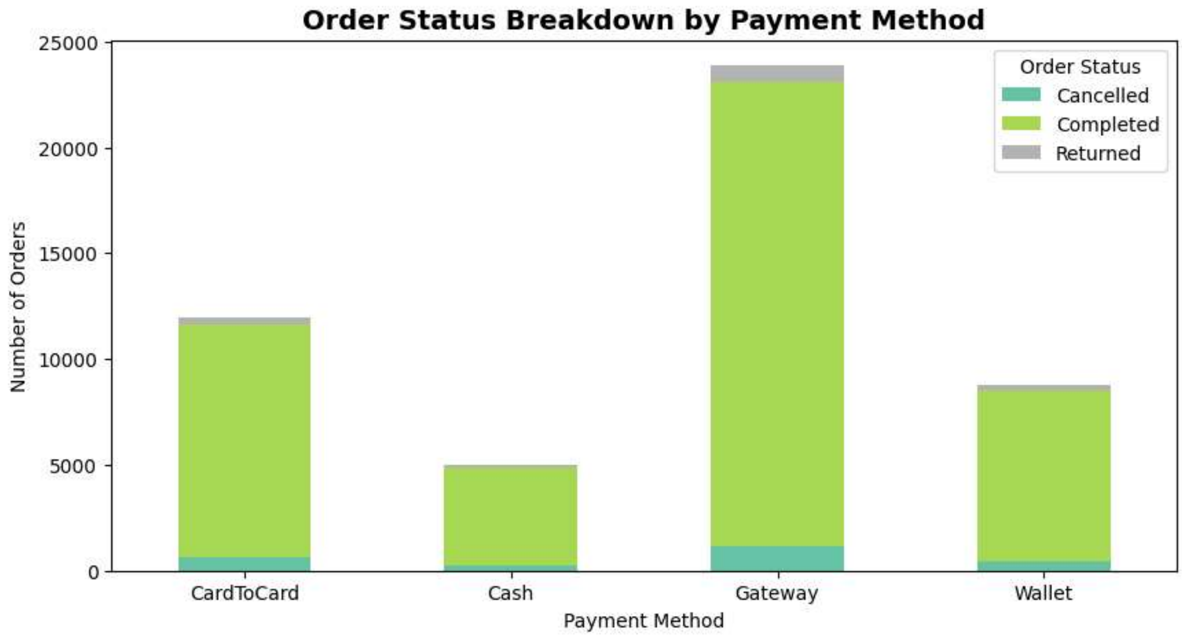

Figure 10: *Order Status Breakdown by Payment Channel (Stacked Bar Chart).*

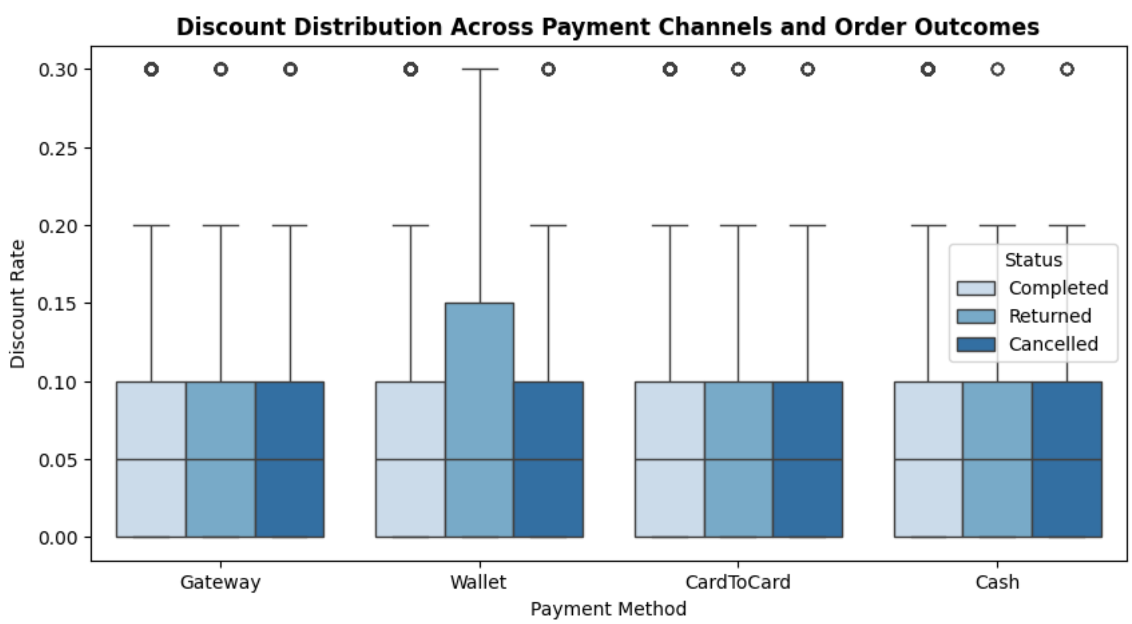

Figure 11: *Discount Distribution Across Payment Modalities and Order Outcomes (Boxplot).*

4. Business Recommendations

1. **Targeted Re-Engagement Campaigns:** Deploy automated promotional discounts to customers entering the high-risk churn window (identified by the SMOTE Random Forest model) within 30 days of inactivity.
2. **Safety Stock Optimization:** Maintain a dynamic safety stock buffer of approximately 12 units above daily projected demand to account for model variance (RMSE: 11.59).
3. **Channel Optimization:** Restructure checkout options or introduce fraud detection improvements for payment channels exhibiting high cancellation/failure rates during Chi-Square friction testing.

5. Conclusion

This project demonstrates an end-to-end analytical framework for e-commerce data engineering, customer behavioral modeling, and operational decision-making. By combining unsupervised clustering, imbalanced machine learning, time-series forecasting, and inferential hypothesis testing, the solution converts raw database records into strategic business assets. Future extensions of this work include integrating gradient boosting models (XGBoost/LightGBM), neural time-series architectures (LSTM/Prophet), and building real-time dashboard integrations.